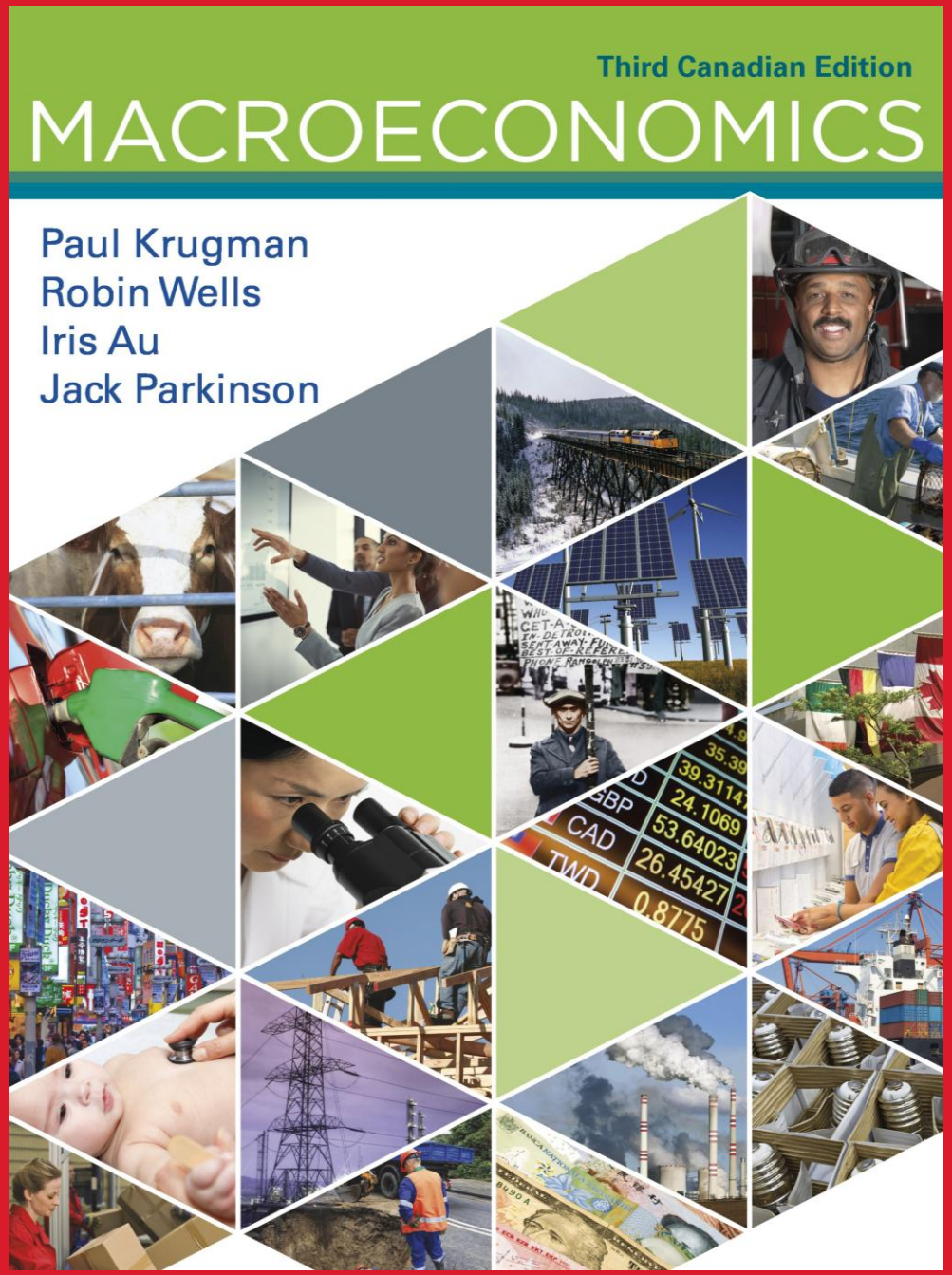
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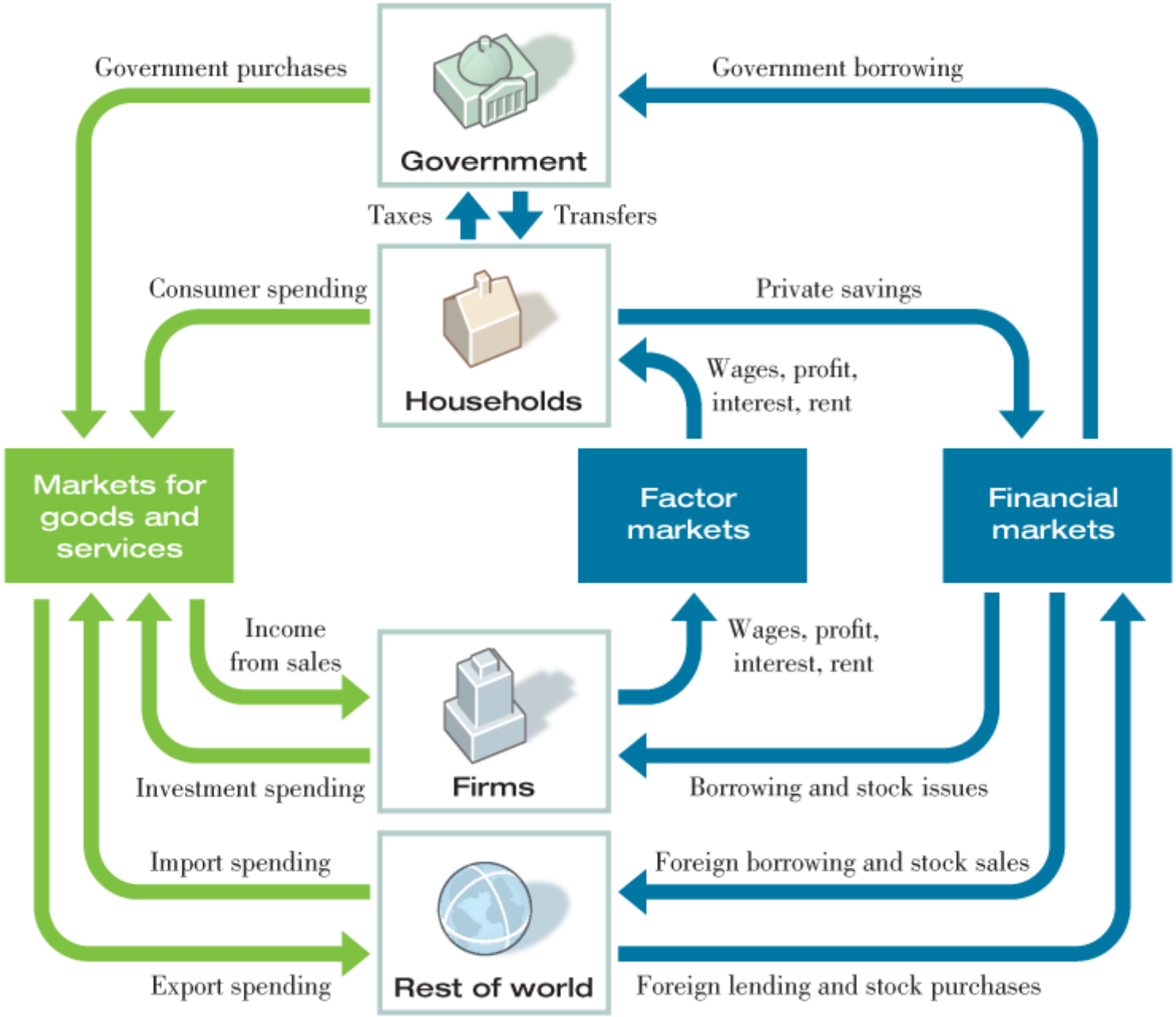
10

SAVINGS, INVESTMENT SPENDING, AND THE FINANCIAL SYSTEM

WHAT YOU WILL LEARN IN THIS CHAPTER

- What is the relationship between savings and investment spending?
- How does the loanable funds market match savers with borrowers?

RECALL CH7 - CIRCULAR FLOW DIAGRAM



MATCHING UP SAVINGS AND INVESTMENT SPENDING

- **Who pays for private investment spending?**
  - Firms' physical capital often financed by other people's savings.
- **Savings–investment spending identity:** savings and investment spending are always equal for the economy.

THE SAVINGS–INVESTMENT SPENDING IDENTITY

Government can also save (or dissave).

- **Budget surplus:** when tax revenue government spending
- **Budget deficit:** when government spending tax revenue
- **Budget balance:** the difference between tax revenue and government spending
- **National savings:** the sum of private savings and the

THE SAVINGS–INVESTMENT SPENDING IDENTITY IN A CLOSED ECONOMY

- Public (Government) savings

$$S_{Public} = T - TR - G$$

- National Savings:

$$S_{National} = S_{Public} + S_{Private}$$

- And since $S = I$

THE DIFFERENT KINDS OF CAPITAL

1. **Physical capital** consists of manufactured resources, such as buildings and machines.
2. **Human capital** is the improvement in the labor force generated by education and knowledge.
3. **Financial capital** is funds from savings that are available for investment spending.

THE SAVINGS–INVESTMENT SPENDING IDENTITY IN AN OPEN ECONOMY

- Inflow of funds : foreign savings that finance domestic investment (capital inflows)
- Outflow of funds: domestic savings that finance foreign investment (capital outflows)

- **Net foreign investment (or NFI) =**

  total outflow of funds - the total inflow of funds

THE SAVINGS–INVESTMENT SPENDING IDENTITY IN AN OPEN ECONOMY

$$NFI = X - IM$$

- If NFI < 0 : Foreigners invest more

- Recall GDP = $C + I + G + X - IM$, or

$$(GDP - C - G) = I + (X - IM)$$

- Since GDP − C − G = national savings,

$$S_{National} = I + NFI$$

THE SAVINGS–INVESTMENT SPENDING IDENTITY IN OPEN ECONOMIES

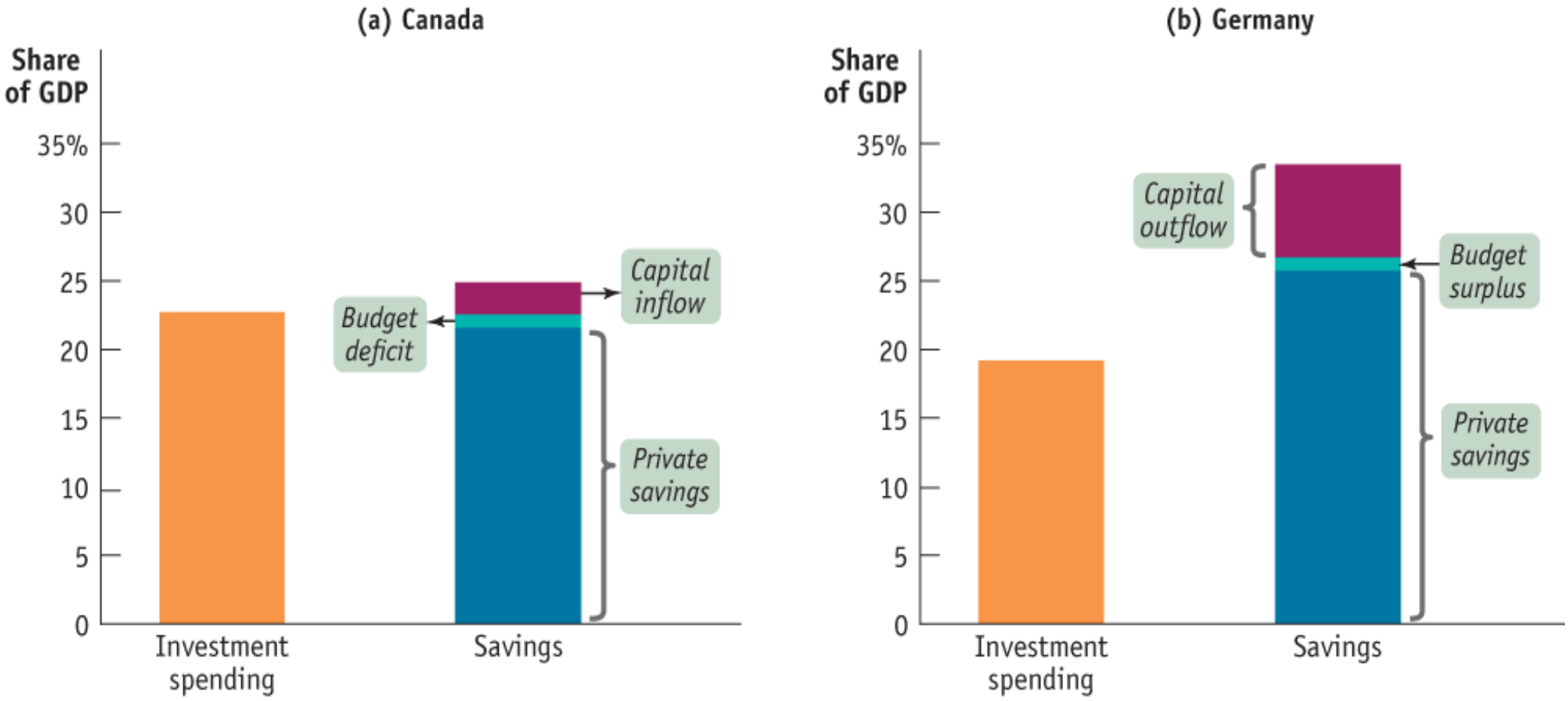


FIGURE 10-1 Krugman/Wells/Au/Parkinson, *Macroeconomics*, 3e, Canadian, © 2018 Worth Publishers
Data from: International Monetary Fund.

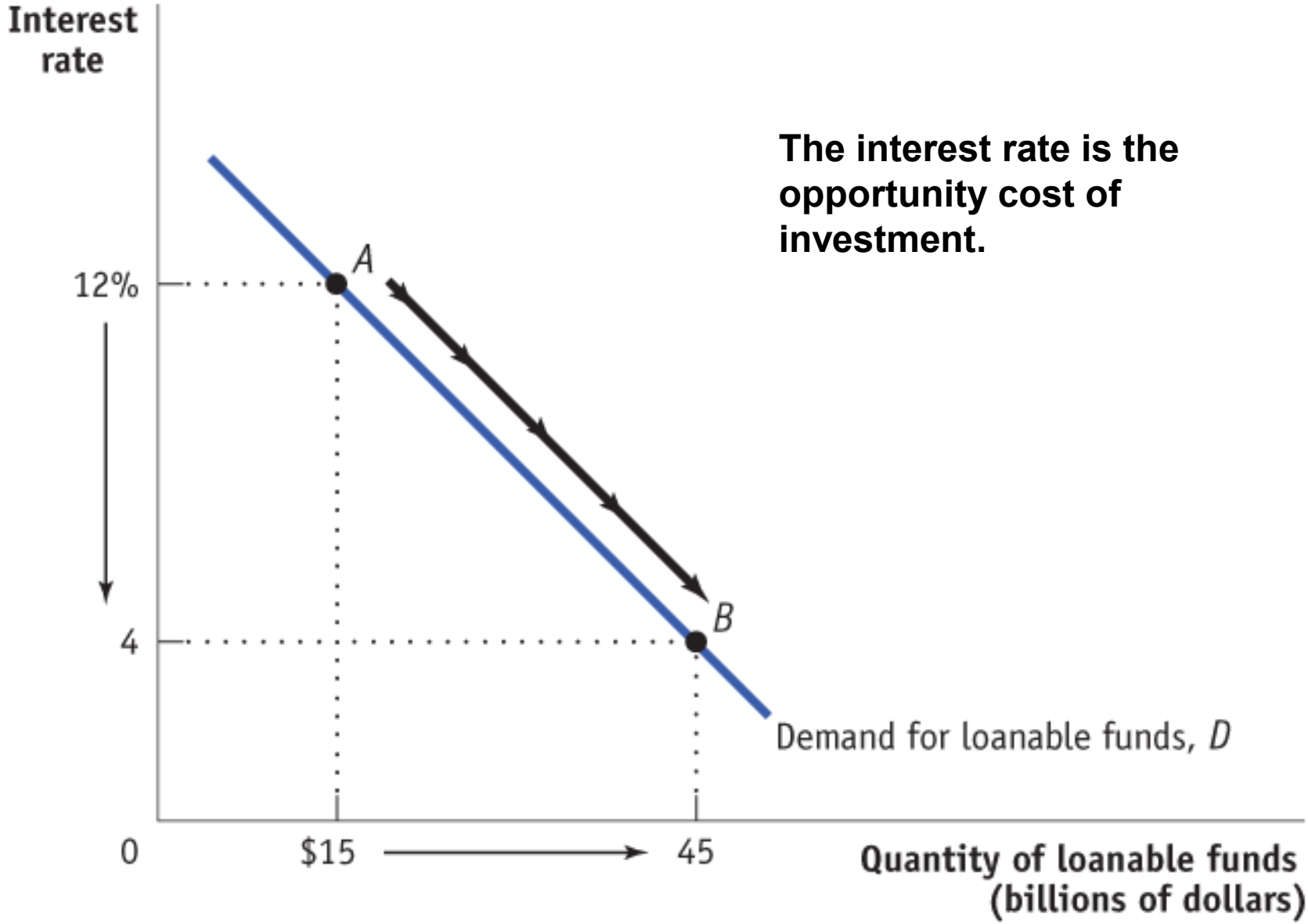
THE MARKET FOR LOANABLE FUNDS

- **Financial markets** channel the savings of households to businesses that want to borrow in order to invest.
- To simplify consider a fictional financial market:
  - **loanable funds market**
  - **price of loans is the (nominal) interest rate**

THE DEMAND FOR LOANABLE FUNDS

- Domestic demand for loanable funds
- Demand for loans by the firms at a given interest rate

THE DEMAND FOR LOANABLE FUNDS



The interest rate is the opportunity cost of investment.

Present Value (PV)

- Firms invest if

  PV of future return on investment …. cost of investment

PV Formula :

$$PV = \frac{FV}{(1 + r)^t}$$

Example :

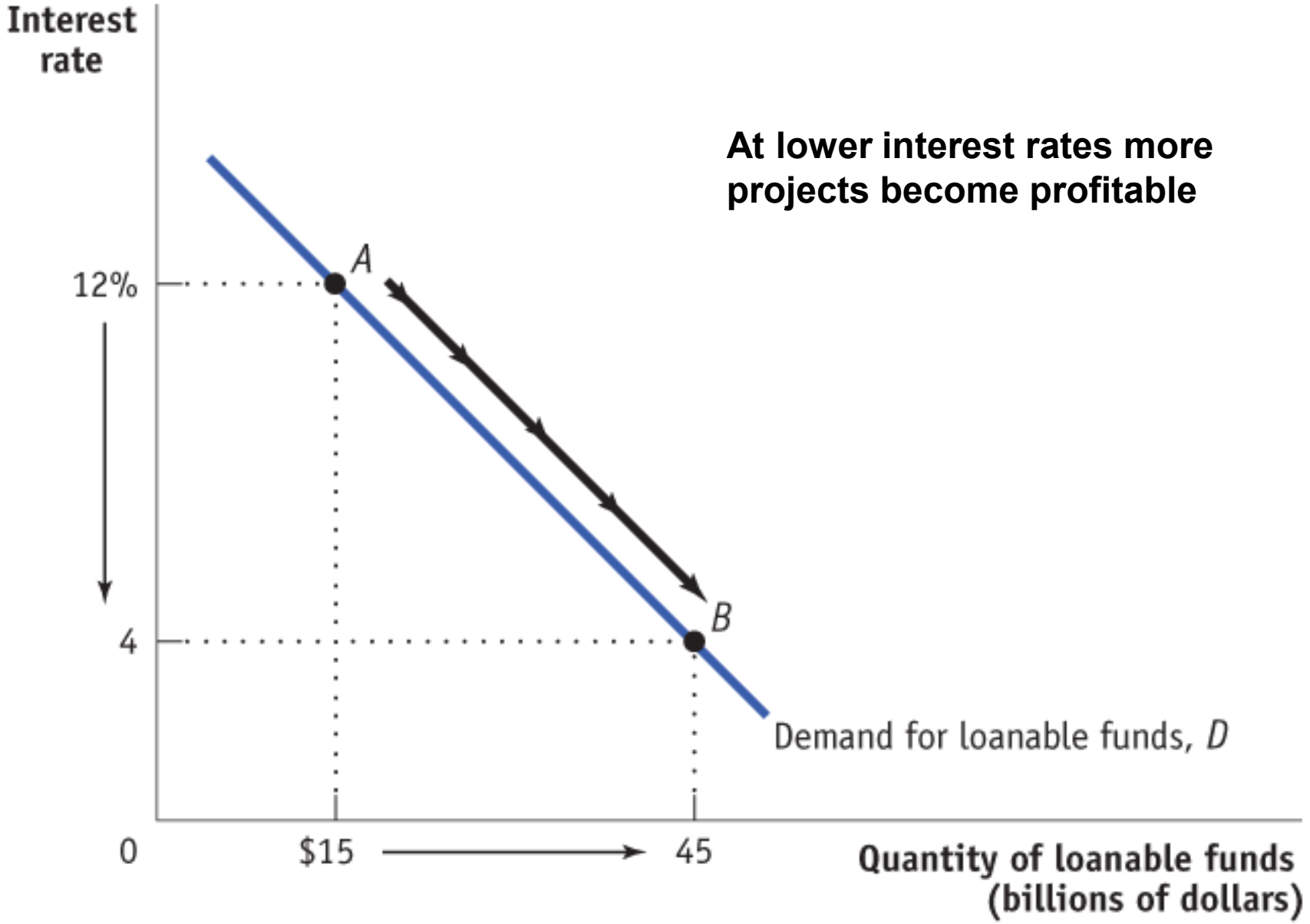
- 2 projects : with one-year return $1000
  - Costs: project 1 : 900 , project 2 : 950

Which of these projects is worth borrowing?

- *Depends on the interest rate.*

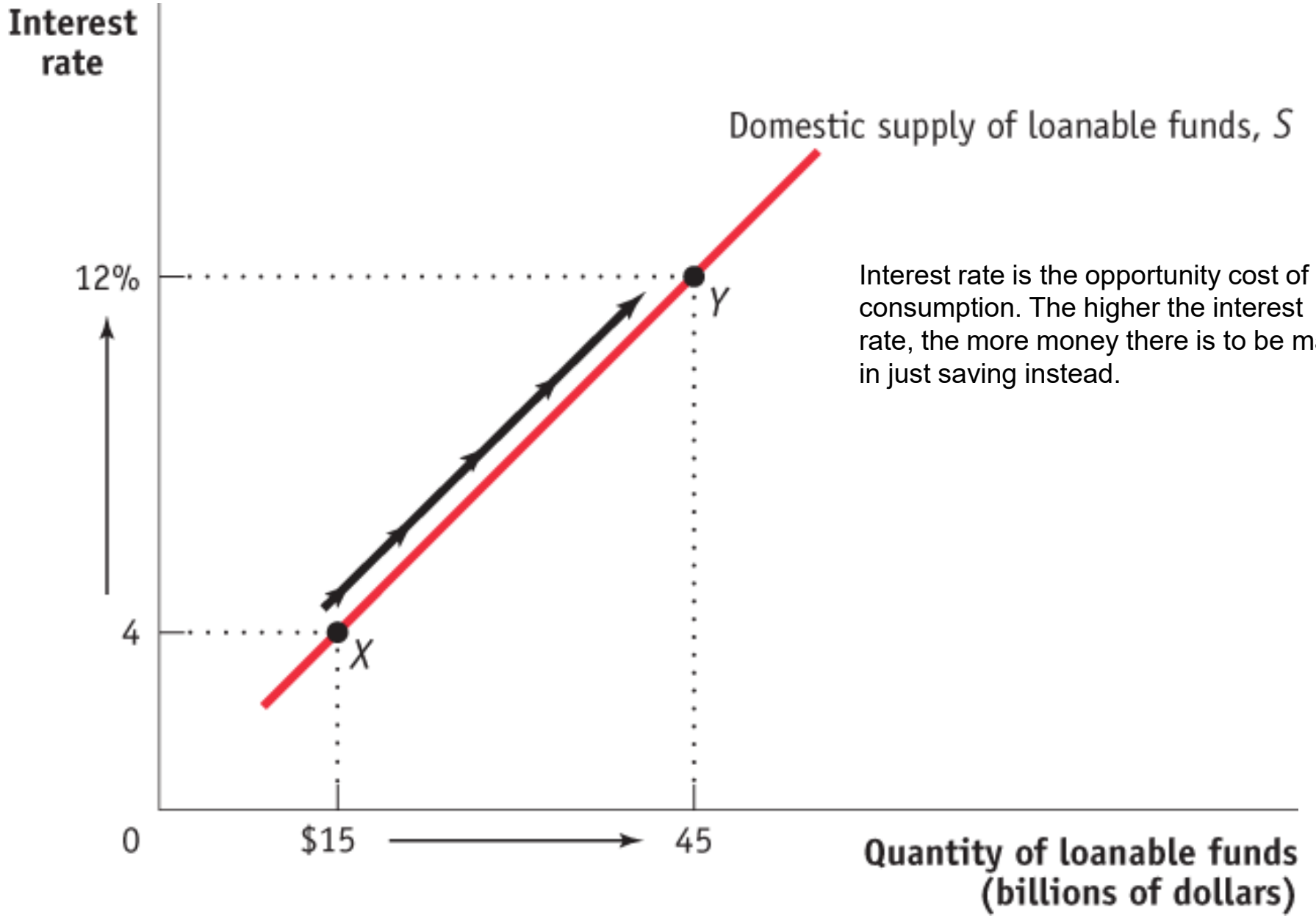
r	FV = 1000	Project 1	Project 2
0.05	PV = 952	Yes	Yes
0.1	PV = 909	Yes	No
0.5	PV = 666	No	no

THE DEMAND FOR LOANABLE FUNDS



At lower interest rates more projects become profitable

THE SUPPLY OF LOANABLE FUNDS

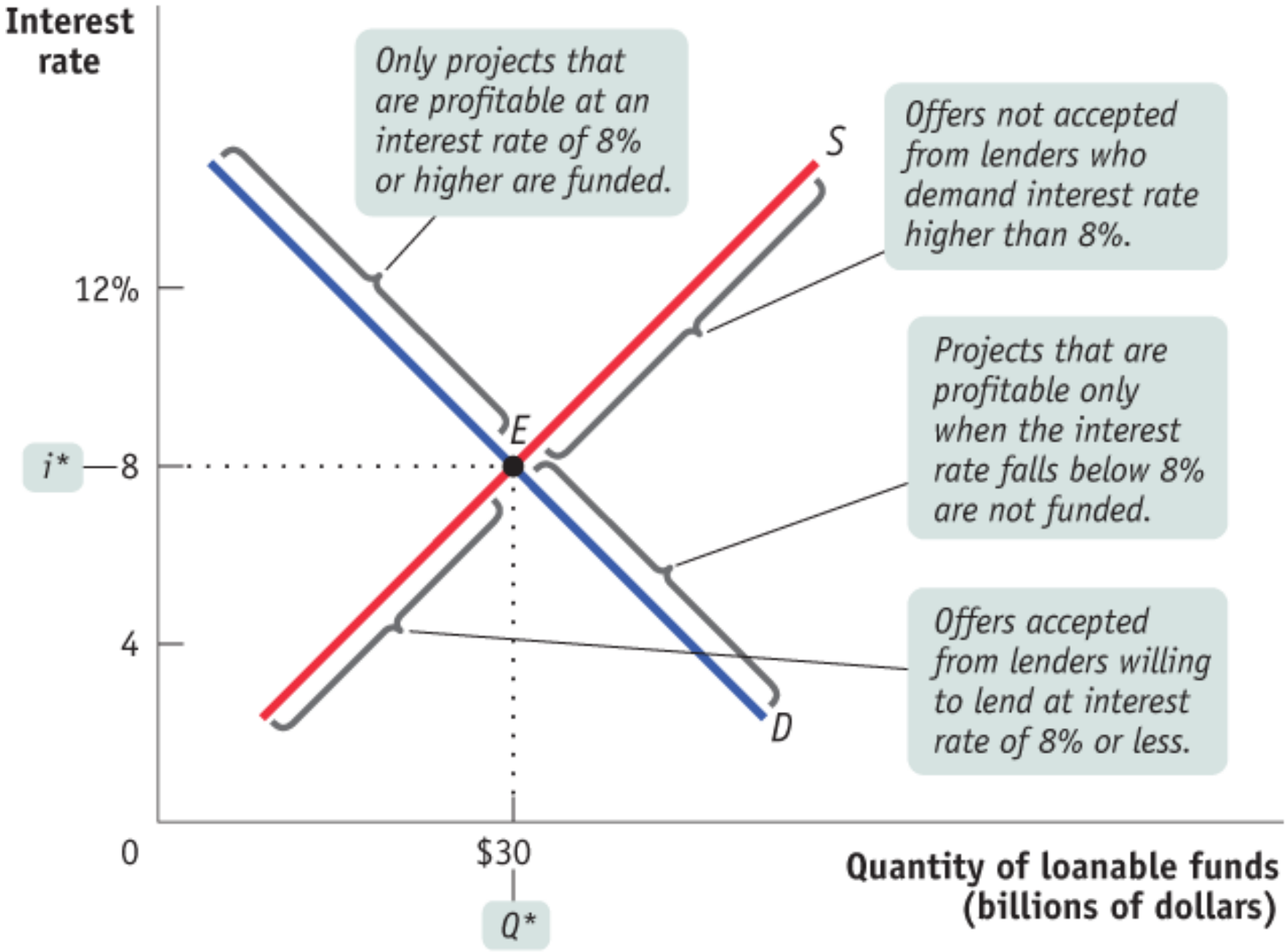


Interest rate is the opportunity cost of consumption. The higher the interest rate, the more money there is to be made in just saving instead.

THE SUPPLY OF LOANABLE FUNDS

- Why does the supply of loanable funds curve slope upward?
  - **More people are willing to lend money when the interest rate is higher.**

THE EQUILIBRIUM INTEREST RATE



THE EQUILIBRIUM INTEREST RATE

- At the equilibrium
  - desired savings = desired investment
  - right projects are funded (those with return at least as high as i*)
  - Right people do savings (those who lend at most at i*)

SHIFTS OF THE DEMAND FOR LOANABLE FUNDS

- Factors that can cause the demand curve for loanable funds to shift:
  - **changes in perceived business opportunities**
    - **Firms might pull back when they are less optimistic about the future**
  - **changes in government policies**
    - **Tax credit for investment**
    - **Ex: Carney implemented a tax credit for foreign investors to incentivize foreign investment**

INCREASE IN THE DEMAND

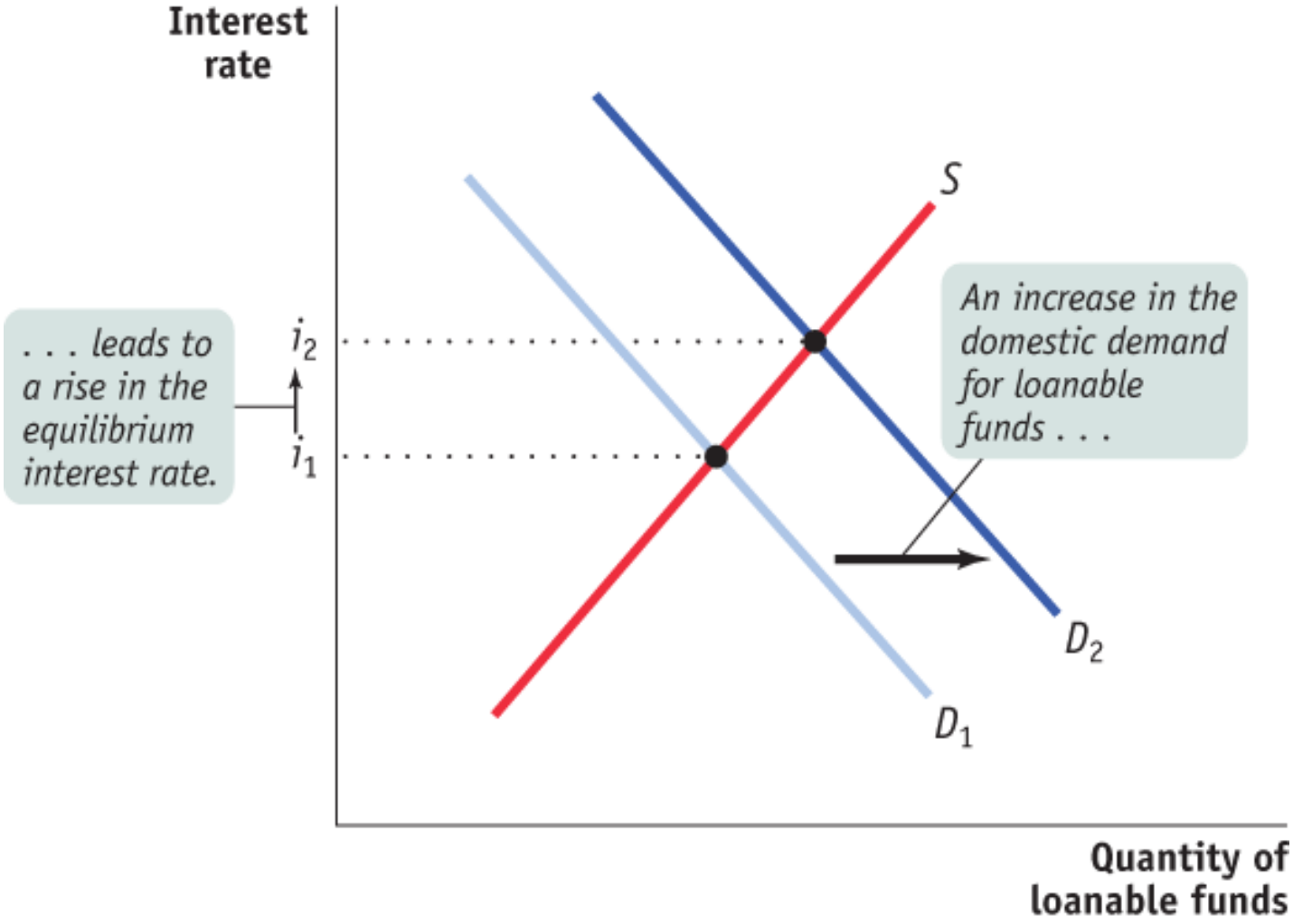


FIGURE 10-5 Krugman/Wells/Au/Parkinson, *Macroeconomics*, 3e, Canadian,
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SHIFTS OF THE SUPPLY OF LOANABLE FUNDS

- Factors that can cause the supply curve for loanable funds to shift:
  - **changes in private savings behavior**
    - **Households may speculatively worry about the future and save more as they realize they are less secure**
    - **HH's may be bullish on the future and borrow and spend more as they realize they are more secure**
  - **changes in government budget balance**
    - $S_{National} = S_{Public} + S_{Private}$

EFFECTS OF BUDGET DEFICIT

- **Crowding out** occurs when a government budget deficit drives up the interest rate and leads to reduced investment spending.

A DECREASE IN THE S OF LOANABLE FUNDS

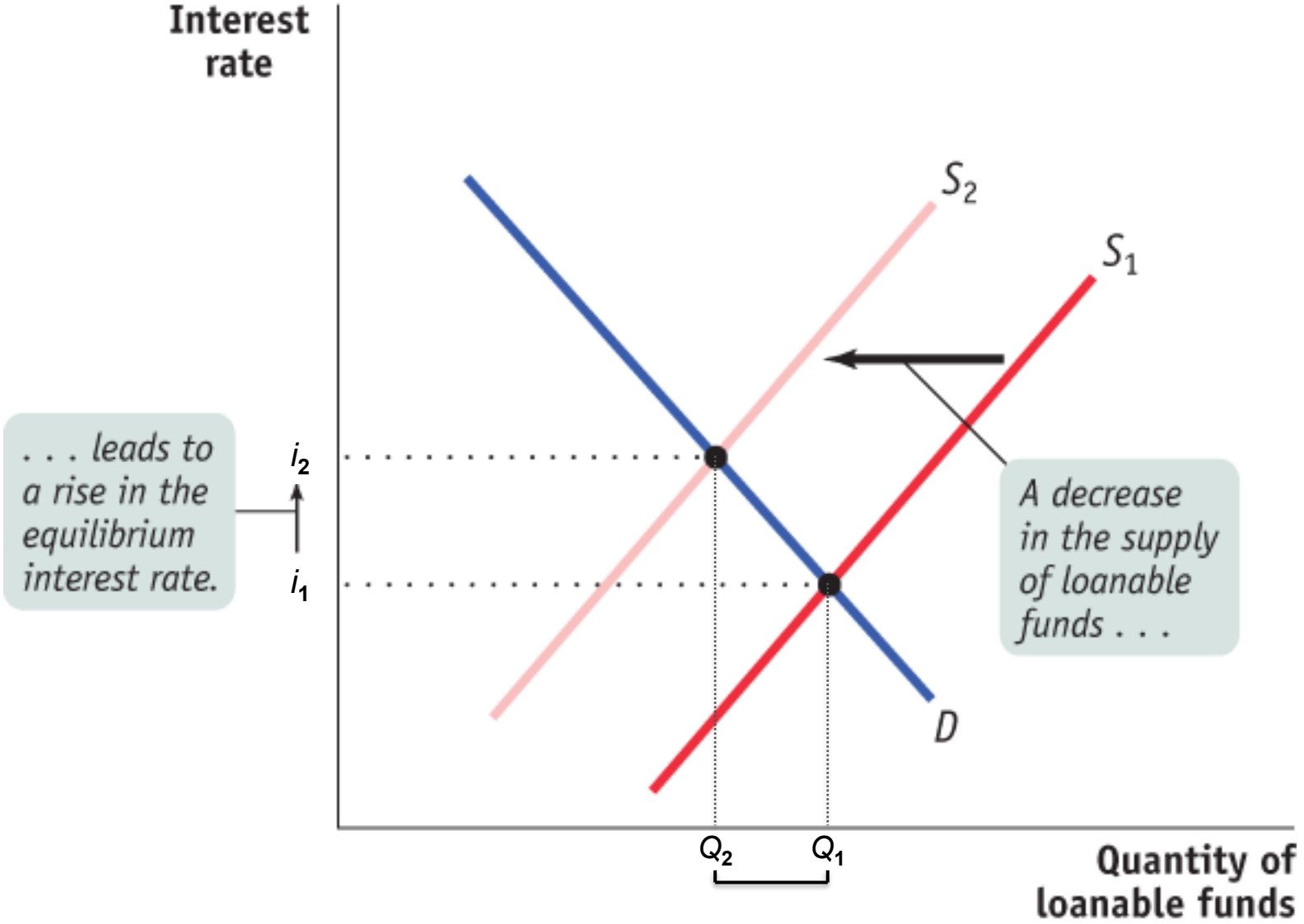
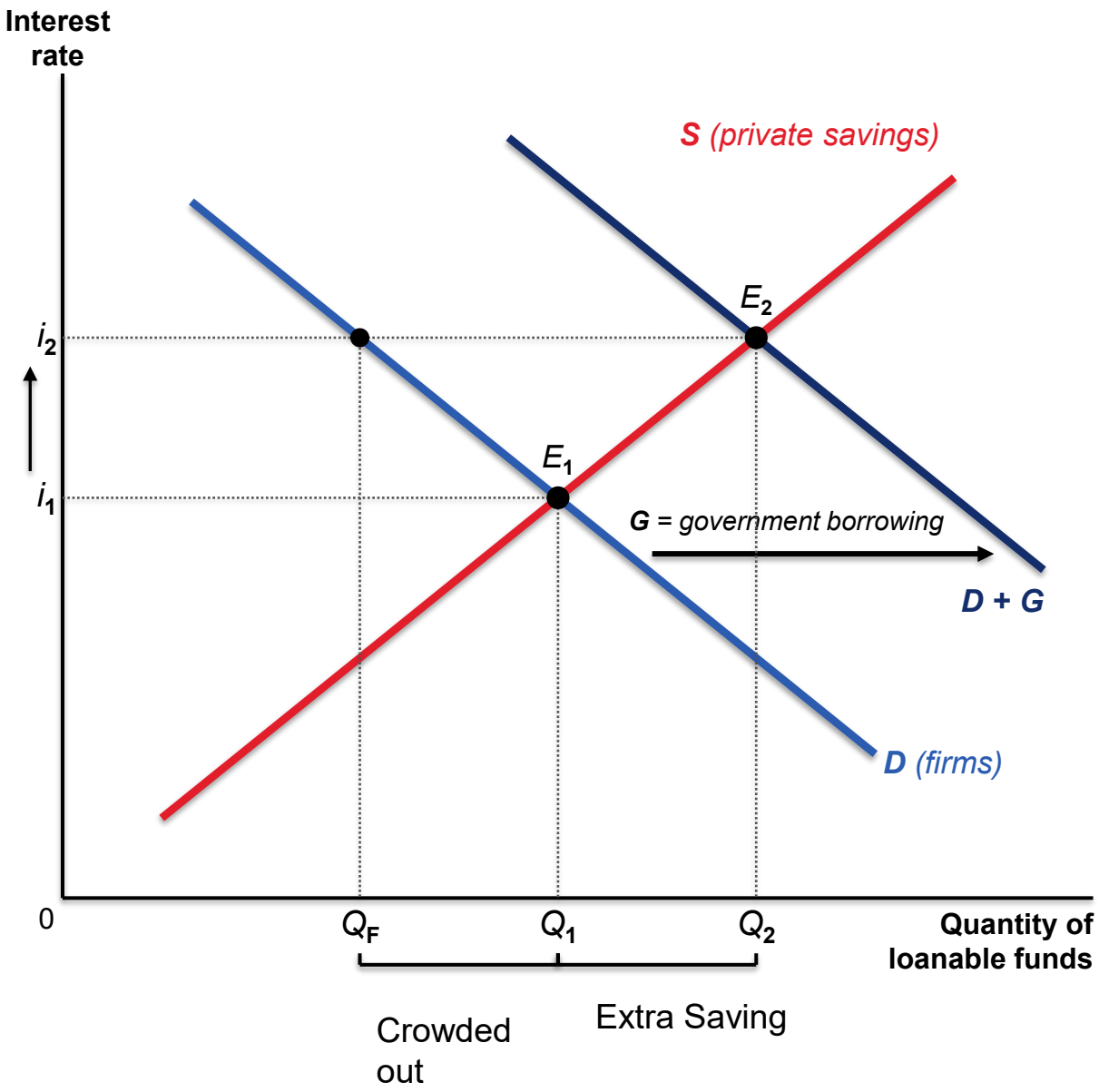


FIGURE 10-6 Krugman/Wells/Au/Parkinson, *Macroeconomics*, 3e, Canadian,

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ANOTHER VIEW: DEFICIT SHIFTS DEMAND



The government borrows in the same market as firms.

1. Demand shifts right: $D \rightarrow$ D + G

2. Interest rate rises: i1 -> i2

3. Firms borrow less: $Q_1 \rightarrow$ ……
This drop is called ……

4. Higher *i* draws out ……
So crowding out is partial: *G* = …… + ……

Same result as the previous slide: ……
Here *S* = private savings only. On the previous slide *S* = national savings, so the deficit was already inside *S*. Shift one curve, never both.

GLOBAL LOANABLE FUNDS

- There is an incentive for capital to flow to where the returns are higher
- A **global loanable funds market can** equalize interest rates across countries.

GLOBAL LOANABLE FUNDS

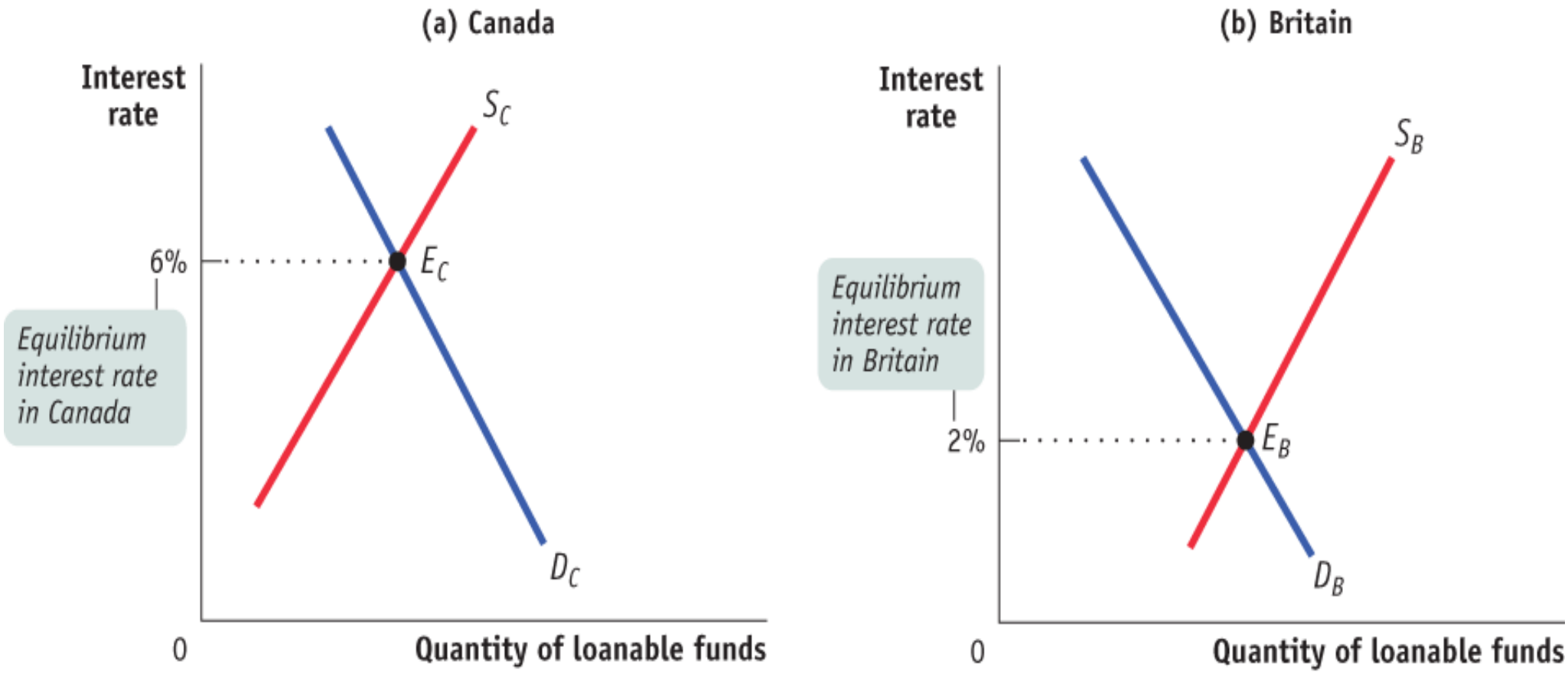


FIGURE 10-7 Krugman/Wells/Au/Parkinson, *Macroeconomics,* 3e, Canadian, © 2018 Worth Publishers

EQUILIBRIUM IN THE GLOBAL MARKET

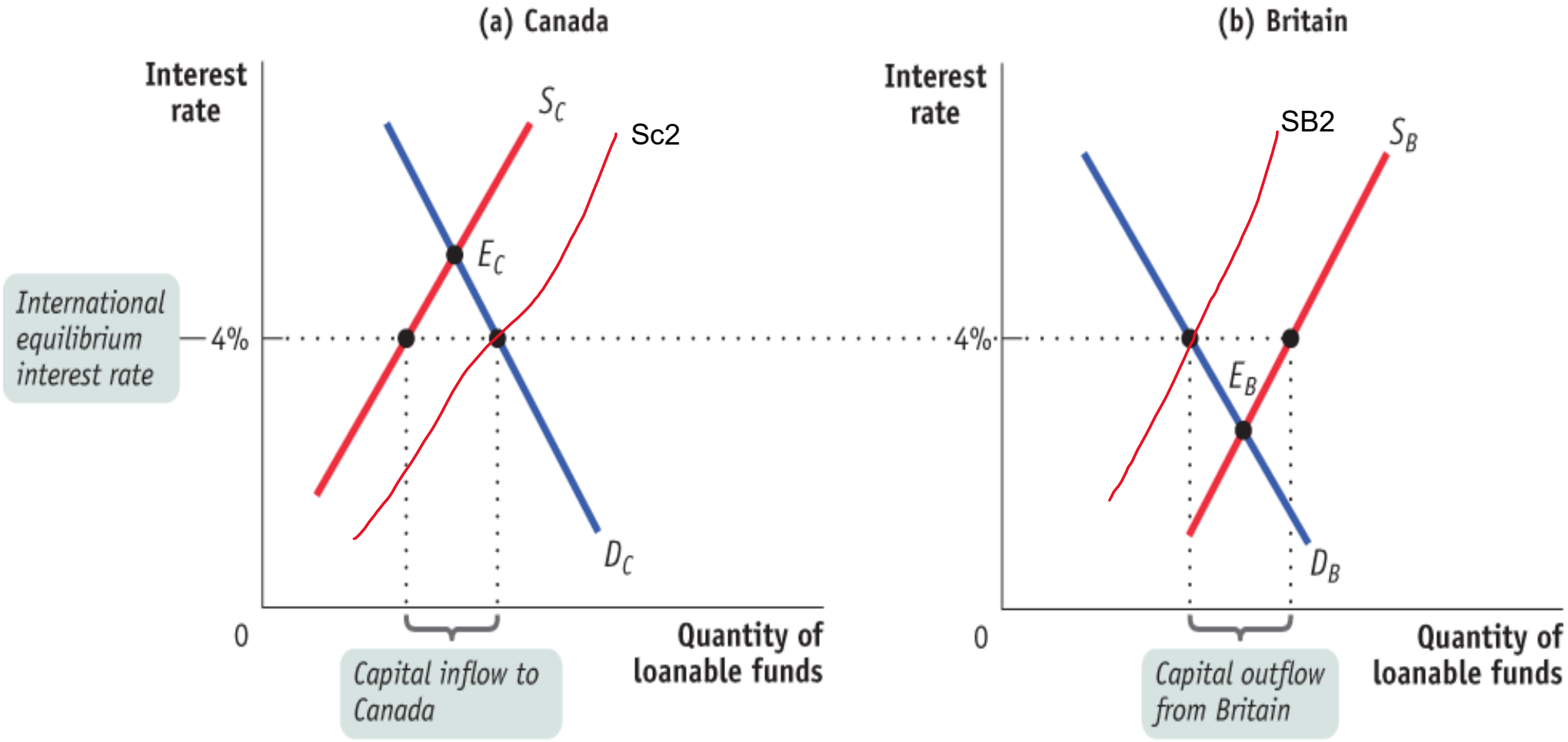


FIGURE 10-8 Krugman/Wells/Au/Parkinson, *Macroeconomics*, 3e, Canadian, © 2018 Worth Publishers

Data

INFLATION AND INTEREST RATES

- **Shifts in supply or demand of loanable funds changes the interest rate.**
- Major changes in interest rates have been driven by many factors, including:
  - changes in government policy.
  - technological innovations that created new investment opportunities.
- And, most important, **expectations about future inflation**

INFLATION AND INTEREST RATES

- **Real interest rate = nominal interest rate – inflation rate.**
- The **true cost of borrowing (and payoff to lending)** is the **real interest rate.**
- But neither lenders nor borrowers know exactly what inflation will be, so loan contracts specify a nominal interest rate

THE FISHER EFFECT

- **Fisher effect:**
  - *expected real interest rate is unaffected by changes in expected future inflation*
  - expected future inflation drives up the nominal interest rate

THE FISHER EFFECT IN GRAPHS

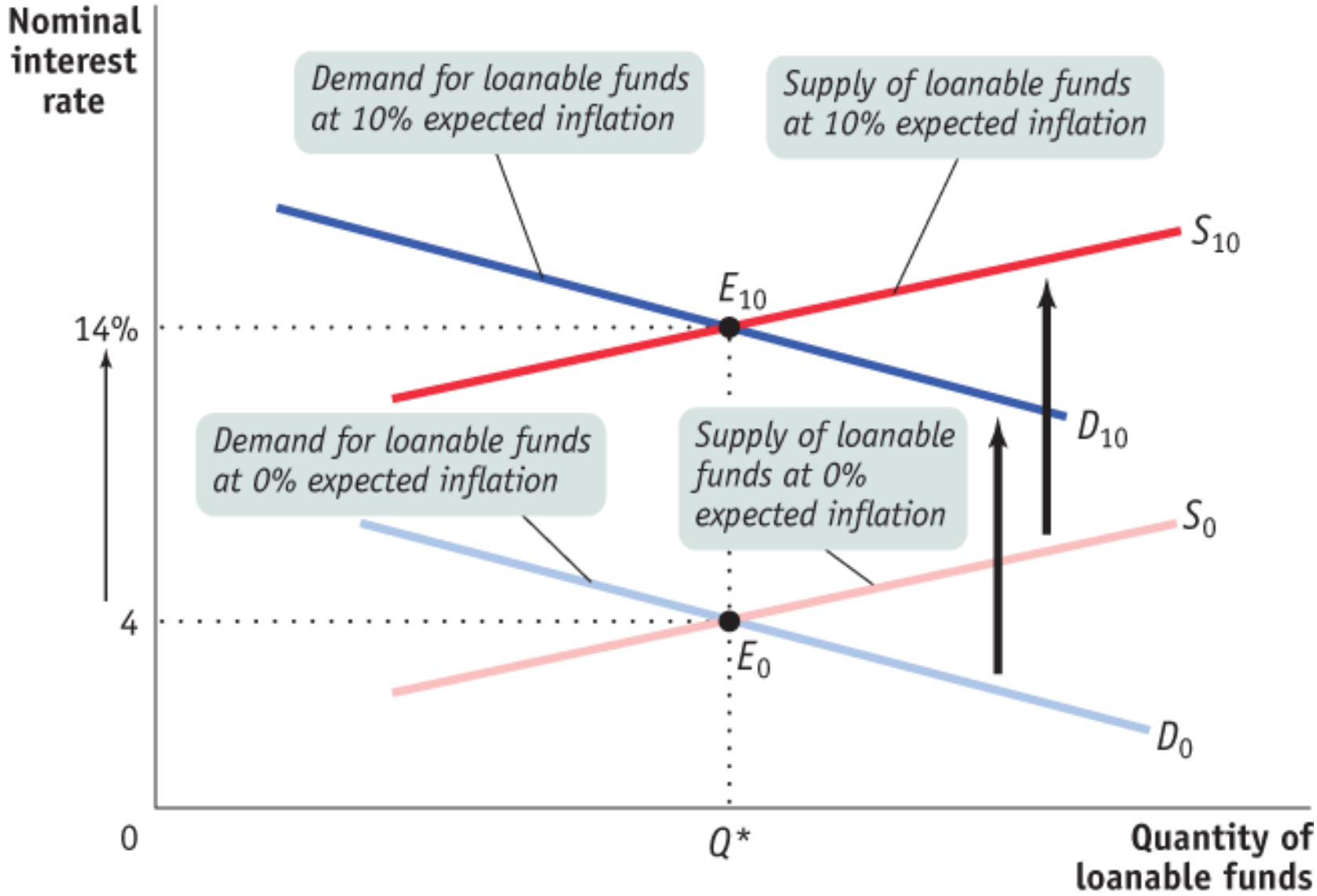


FIGURE 10-9 Krugman/Wells/Au/Parkinson, *Macroeconomics*, 3e, Canadian,

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SIXTY YEARS OF CANADIAN INTEREST RATES

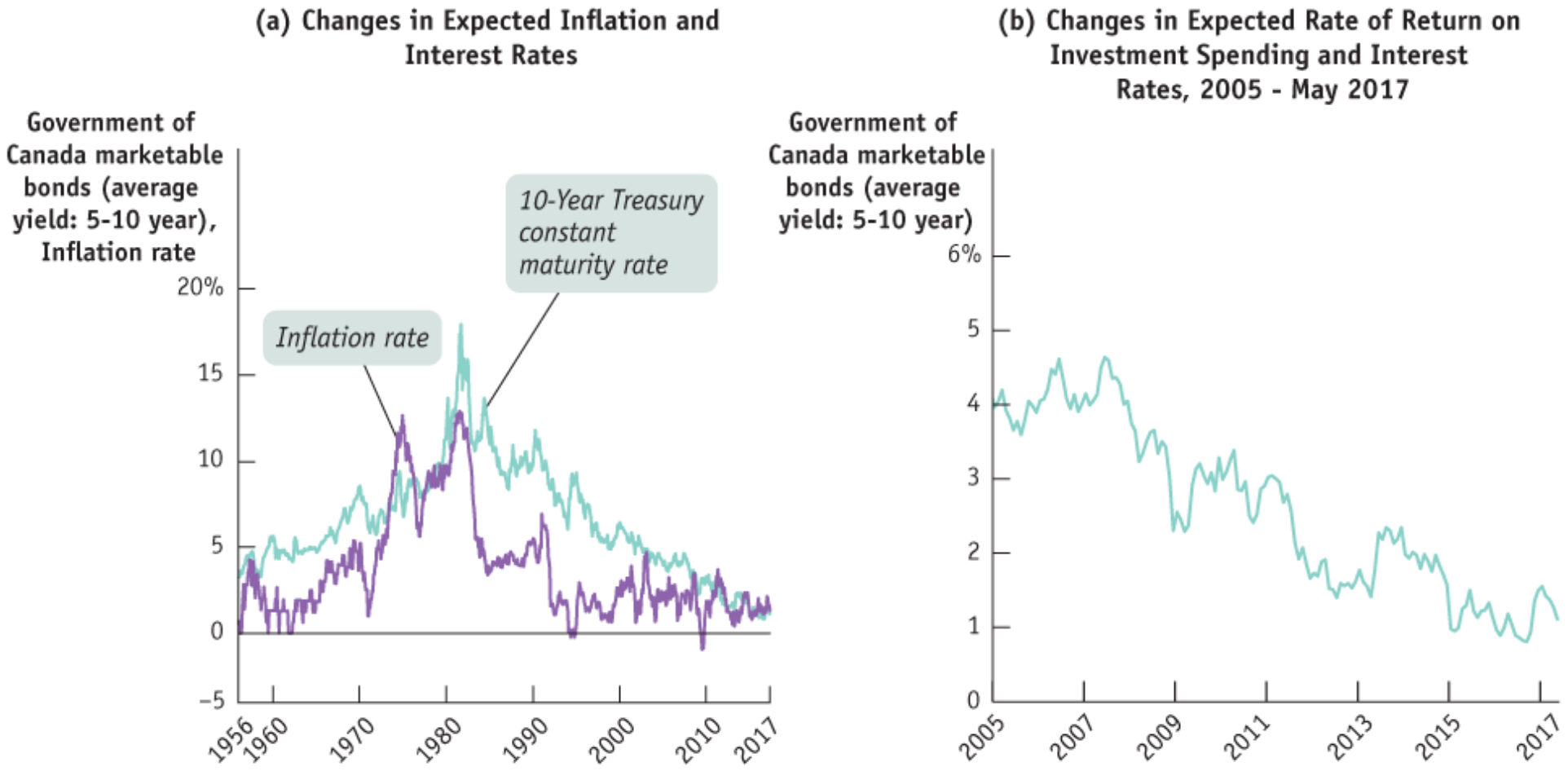
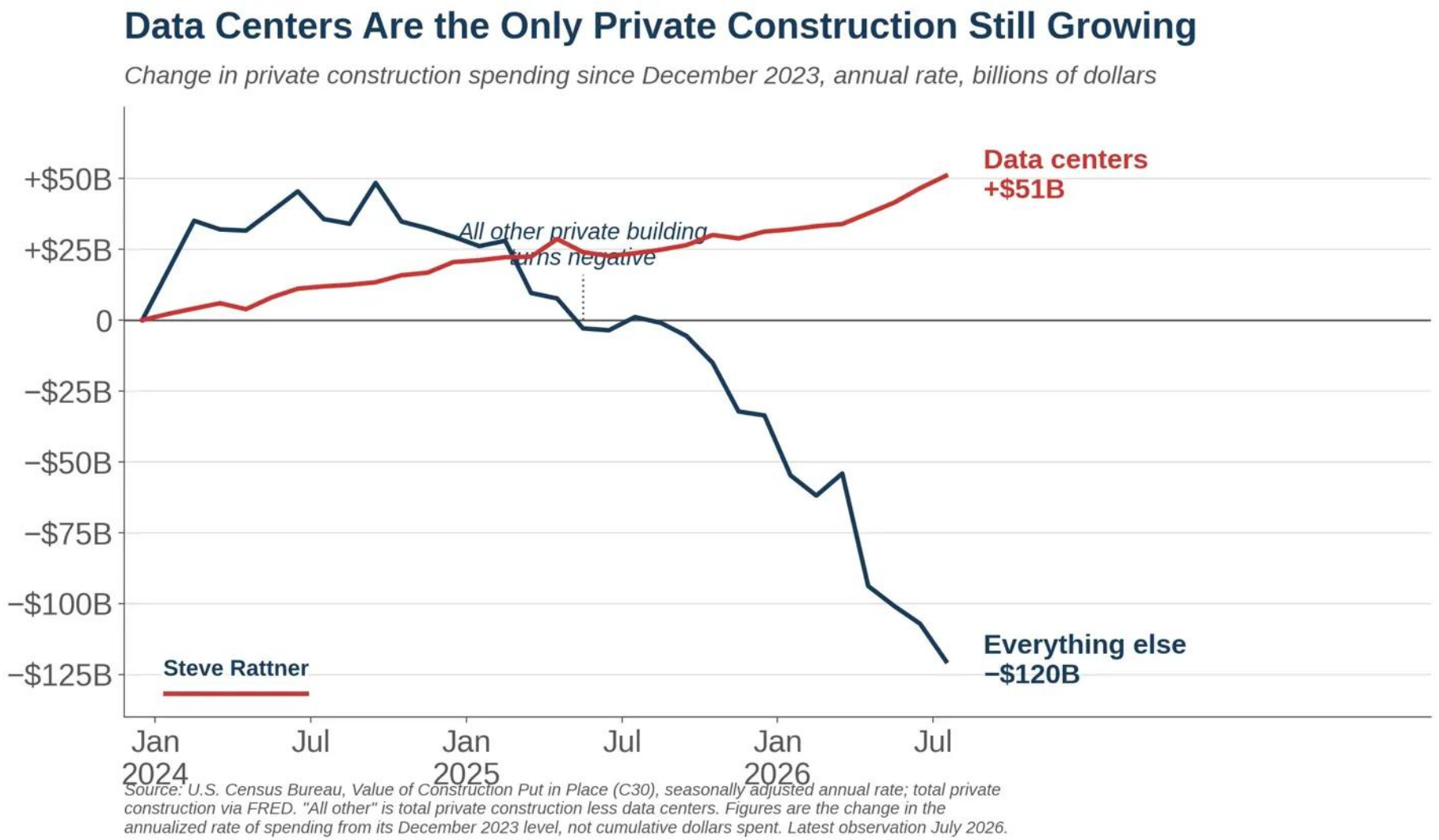


FIGURE 10-10 Krugman/Wells/Au/Parkinson, *Macroeconomics*, 3e, Canadian, © 2018 Worth Publishers

WHY ARE INTEREST RATES HIGH RIGHT NOW?

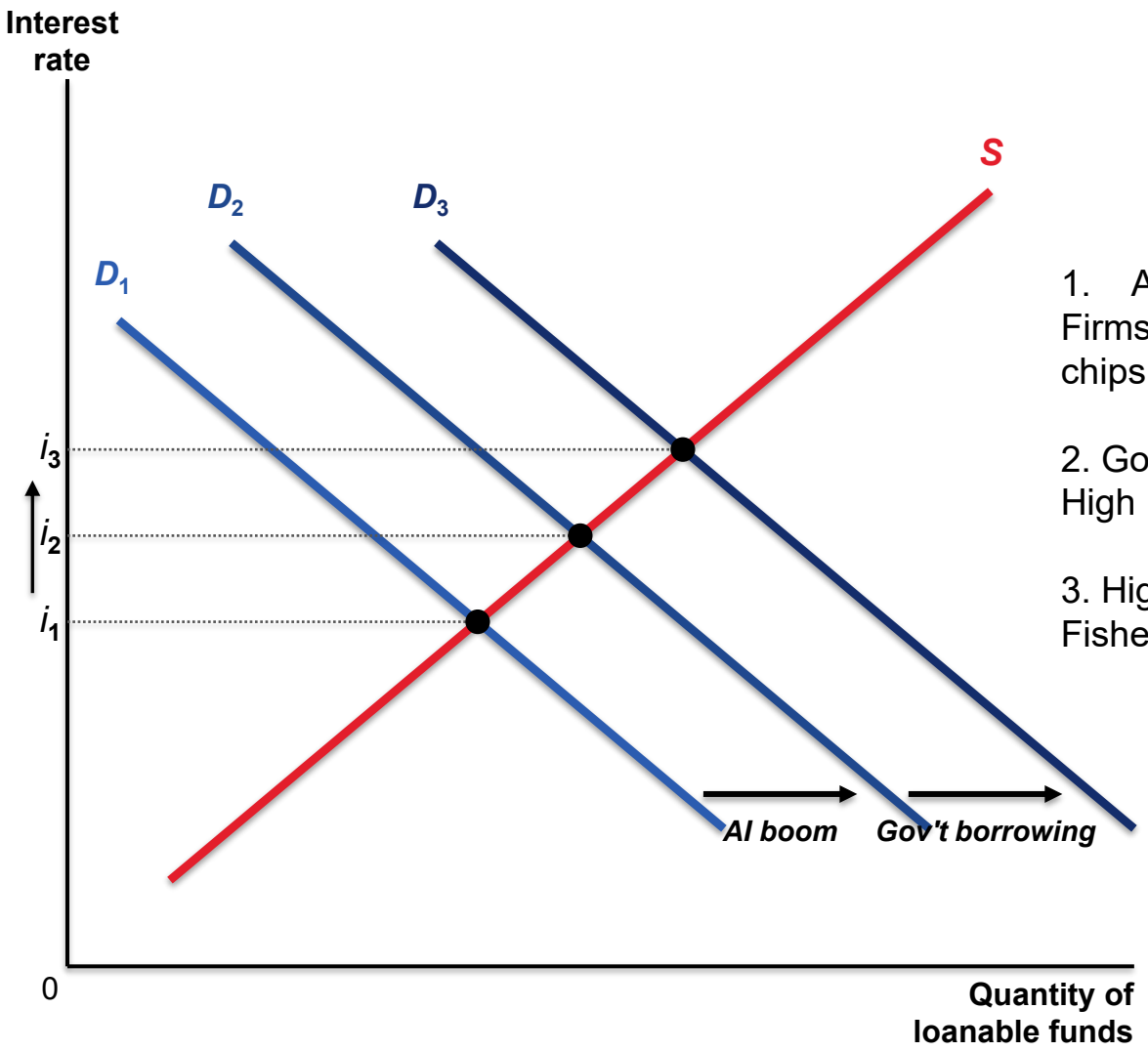
- **Oct 2026**
- U.S. 10-yr bond ≈ 5.1%, highest in almost 20 years.
- **Why?**
  - **Higher expected inflation**
    - High expected inflation due to geopolitical turmoil
    - Fisher effect: nominal rates rise with expected inflation
  - **Government borrowing**
    - US deficit is approx. 6% of GDP; govt. competes with private firms for funding
  - **AI investment boom**
    - Firmst expect high returns on data centres and chips
    - Firms use massive amounts of leverage to fund their investment, this spike in demand moves interest rates up

THE AI BOOM CROWDS OUT OTHER INVESTMENT



Demand for funds ↑ → *i* ↑ → other interest-sensitive investment …… This is called ……

WHY ARE INTEREST RATES HIGH RIGHT NOW?



1. AI investment boom
Firms expect high returns on data centres and chips, demand for loanable funds shifts right

2. Govt. Borrowing
High US deficit

3. Higher expected future inflation
Fisher effect: rates rise with inflation